

Expt. No. : 16

Date :

IMPLEMENT PIE CHART USING MATPLOTLIB

AIM

To write a python program to draw the pie chart using matplotlib.

SOFTWARE REQUIRED

Python 3.14

THEORY

A pie chart is a circular statistical graphic that is divided into slices to illustrate numerical proportions. Each slice of the pie represents a category's contribution to the whole, making it easy to visualize relative sizes and percentages. Pie charts are commonly used in business and data analysis to represent data distributions, market shares, or any other categorical data.

Key Features of Pie Charts

Proportional Representation: Each slice's size is proportional to the quantity it represents. For example, if a category makes up 25% of the total, its corresponding slice will occupy a quarter of the pie.

Visual Impact: Pie charts provide a quick and intuitive way to grasp the distribution of data. They are particularly effective when the number of categories is limited (ideally fewer than six).

Color Coding: Different colors can be assigned to each slice to enhance clarity and distinction between categories.

- **Percentage Labels:** Pie charts can display the percentage of each category, allowing viewers to quickly understand the relative sizes of the slices.

PROBLEM

1. Build a Python application to read and display real-time temperature data continuously and identify values exceeding a specified limit.
2. Build a Python program using Matplotlib to represent the percentage contribution of different product categories to total sales using a pie chart.
3. Write a Python program using Matplotlib to create a pie chart showing the distribution of students based on their favourite programming languages: Python (40), Java (25), C++ (20), and JavaScript (15).

ALGORITHM

STEP 1: Start the program

STEP 2: Import matplotlib.pyplot as plt.

STEP 3: Define labels, sizes, colors, and optionally explode.

STEP 4: Initialize the figure with plt.figure(figsize=(8, 6)).

STEP 5: Generate the pie chart using plt.pie() with the prepared data.

STEP 6: Ensure the pie chart is circular with plt.axis('equal').

STEP 7: Set a title for the chart using plt.title('Your Title Here').

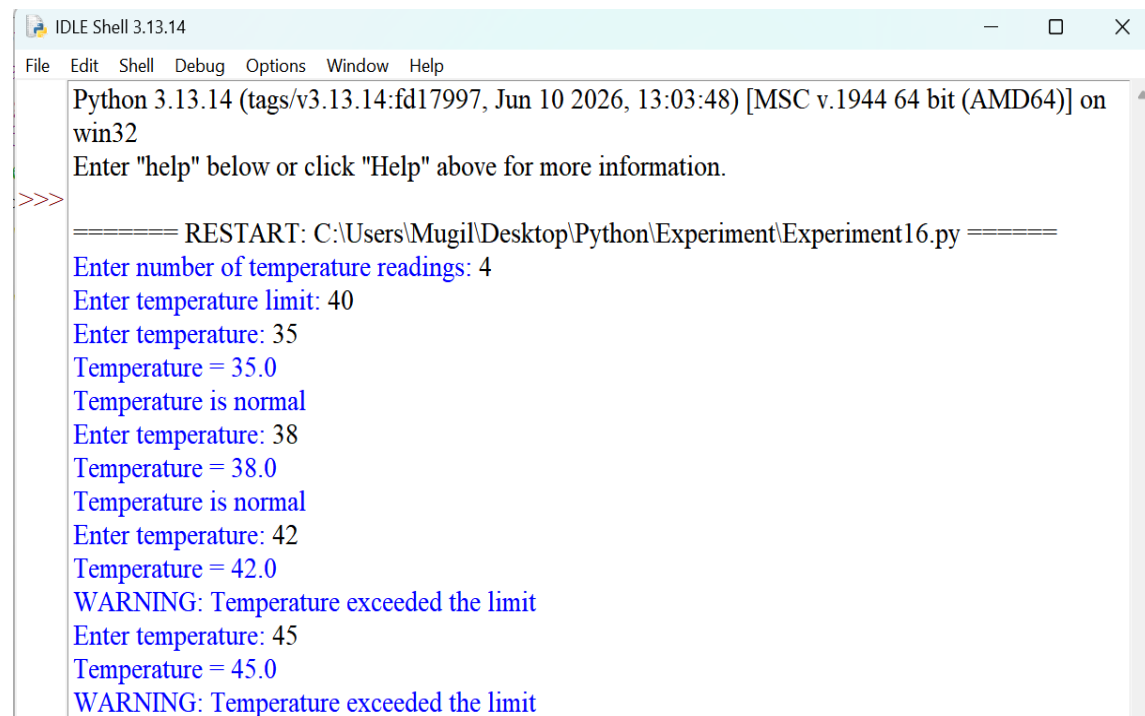
STEP 8: Render the pie chart with plt.show().

STEP 9: Stop the program

PROGRAM CODING

```
n = int(input("Enter number of temperature readings: "))
limit = float(input("Enter temperature limit: "))
for i in range(n):
    temp = float(input("Enter temperature: "))
    print("Temperature =", temp)
    if temp > limit:
        print("WARNING: Temperature exceeded the limit")
    else:
        print("Temperature is normal")
```

OUTCOMES OF PROGRAM CODING



```
IDLE Shell 3.13.14
File Edit Shell Debug Options Window Help
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\Mugil\Desktop\Python\Experiment\Experiment16.py =====
Enter number of temperature readings: 4
Enter temperature limit: 40
Enter temperature: 35
Temperature = 35.0
Temperature is normal
Enter temperature: 38
Temperature = 38.0
Temperature is normal
Enter temperature: 42
Temperature = 42.0
WARNING: Temperature exceeded the limit
Enter temperature: 45
Temperature = 45.0
WARNING: Temperature exceeded the limit
```

PROBLEM CODING

```

import matplotlib.pyplot as plt
n = int(input("Enter number of product categories: "))
names = []
sales = []
for i in range(n):
    name = input("Enter product category: ")
    value = float(input("Enter sales: "))
    names.append(name)
    sales.append(value)
plt.pie(sales, labels=names, autopct="%1.1f%%")
plt.title("Product Sales")
plt.show()

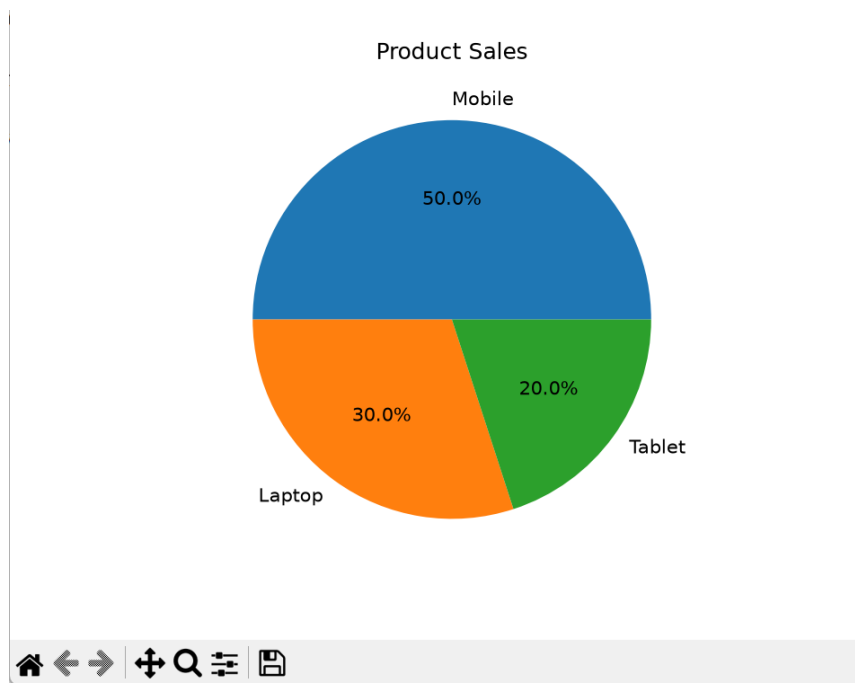
```

OUTCOMES OF PROGRAM CODING

```

Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on
win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\Mugil\Desktop\Python\Problem\experiment16.py =====
Enter number of product categories: 3
Enter product category: Mobile
Enter sales: 50000
Enter product category: Laptop
Enter sales: 30000
Enter product category: Tablet
Enter sales: 20000

```

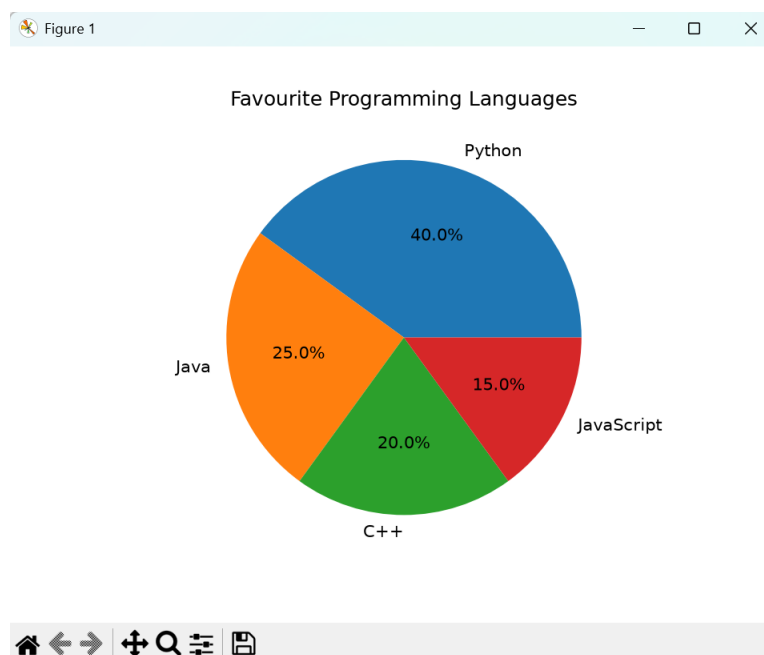


PROBLEM CODING

```
import matplotlib.pyplot as plt
python = float(input("Enter Python percentage: "))
java = float(input("Enter Java percentage: "))
cpp = float(input("Enter C++ percentage: "))
javascript = float(input("Enter JavaScript percentage: "))
languages = ["Python", "Java", "C++", "JavaScript"]
values = [python, java, cpp, javascript]
plt.pie(values, labels=languages, autopct="%1.1f%%")
plt.title("Favourite Programming Languages")
plt.show()
```

OUTCOMES OF PROGRAM CODING

```
Python 3.13.14 (tags/v3.13.14:fd17997, Jun 10 2026, 13:03:48) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
===== RESTART: C:\Users\Mugil\Desktop\Python\Problem1\16.py =====
Enter Python percentage: 40
Enter Java percentage: 25
Enter C++ percentage: 20
Enter JavaScript percentage: 15
```



VIVA QUESTION**1. What is a pie chart, and when would you use it?**

A pie chart is a circular statistical graphic divided into slices to illustrate numerical proportions. It is used to represent the contribution of different categories to a whole, making it effective for displaying data with a limited number of categories (ideally less than six).

2. What library in Python is commonly used to create pie charts?

The matplotlib library, specifically matplotlib.pyplot, is commonly used to create pie charts in Python due to its simplicity and flexibility for data visualization.

3. How do you ensure that the pie chart is circular?

By using plt.axis('equal'), we ensure that the aspect ratio is equal, which makes the pie chart appear as a perfect circle.

4. What does the explode parameter do in the plt.pie() function?

The explode parameter allows specific slices of the pie chart to be separated or "exploded" from the rest of the pie. This is often used to highlight particular categories for emphasis.

5. How can you display percentage values on the pie slices?

By setting the autopct parameter in the plt.pie() function, you can format the percentage values displayed on each slice, using a string like '%1.1f%%' for one decimal place.

6. Why is it important to limit the number of slices in a pie chart?

Limiting the number of slices (ideally to fewer than six) helps prevent clutter and confusion, allowing viewers to easily interpret and understand the data being represented.

7. How do you add a title to the pie chart?

Add a title by using the plt.title('Your Title Here') function before displaying the chart.

8. Can you modify the colors of the pie chart slices? How?

Yes, customizing the colors of the pie chart slices by providing a list of color values in the colors parameter when calling the plt.pie() function.

9. What types of data are best suited for pie charts?

Pie charts are best suited for categorical data with a limited number of categories where you want to show the relative size of each category compared to the whole. Examples include market share distribution, survey results, or budget allocations.

RESULT:

Thus, the Python program successfully implemented a pie chart using Matplotlib to represent the given categorical data. The data was divided into appropriate sectors, and the chart was displayed and verified successfully.

INFERENCE:

The experiment demonstrates that a pie chart effectively represents the proportion and percentage contribution of different categories within a dataset. It provides a clear visual comparison of part-to-whole relationships, supporting easy data interpretation and analysis.

Assessment	Max. Marks	Marks Obtained
Conduction of Practical	15	
Report	5	
Viva	5	
Total Marks	25	
Signature of Lab In-Charge		

